



OPEN A temperature- and impedance-aware LSTM–PINN framework for physically consistent battery SOH prediction

Prajakth N. Kumar, Praneeth R. Upadhyaya, S. Nischay, G. Pavan Kumar, Shobana T. S. & Rashmi K. B.

This paper introduces a physics-informed neural network (PINN) and long short-term memory (LSTM) prediction system based on an offline and laboratory-supervised prediction of the state of health (SOH) of a lithium-ion battery. The model proposed combines an electrochemical degradation concept and temporal sequence learning, thus allowing the network to predict SOH development during early-life pathways and conserve consistency with the trends of physical ageing. Time dependencies of degradation dynamics are represented by the LSTM component, and structures the required behavioral limitations imposed by the physics-informed head by Arrhenius-type temperature acceleration, impedance-growth effects as well as monotonic degradation kinetics. The framework uses an adaptive degradation-rate scaling mechanism that integrates explicitly the real-time increase of impedance with thermal kinetics, this enables it to adapt to the changing operating conditions. These constraints lessen dependence on the entirely empirical fit and prevent non-physical outputs so often found in the unconstrained data-driven models. Cycling data evaluation shows that with the hybrid LSTM–PINN design, the root-mean-square error and mean absolute error of SOH trajectory prediction are at 1.01% and 0.6% respectively, and shows physical plausibility exists in conditions of different temperatures and impedances. The paper also emphasizes how the incorporation of the electrochemical ageing laws in the sequence-learning framework can be used to increase the stability of long-horizons and increase the interpretability of battery health prognostics.

Keywords Physics-informed neural networks (PINN), Lithium-ion batteries, State of health (SOH), Electrochemical degradation, Arrhenius temperature dependence, Impedance modeling, Long short term memory (LSTM)

Lithium-ion batteries are very important to electric vehicles, grid-scale storage, and portable electronics, but still, they undergo unavoidable degradation such as capacity fade, impedance rise, and loss of active lithium. Hence accurate estimation of state of health (SOH) is essential for safe and reliable operation. While data-driven deep learning models especially LSTMs capture non-linear temporal degradation patterns effectively^{1,2}, they often require large datasets and may generate non-physical behaviours such as SOH recovery or unrealistic resistance drops. Conversely, physics-based electrochemical models ensure interpretability, but these suffer from high computational cost and poor real-time scalability^{3,4}.

Physics-informed neural networks (PINNs) have emerged as a promising hybrid solution, using electrochemical, thermal or kinetic constraints into the learning process to ensure physical consistency even with limited data. Some recent works demonstrate success in SOH estimation^{5–8}, multi-state co-estimation^{9–11}, electrode-level diagnostics¹², small-sample learning⁷, and degradation-aware modeling^{13,14}. Extensions such as PINN–LSTM hybrids¹⁵, PINN–SPM solvers¹⁴, Bayesian PINNs¹⁶, fractional-order PINNs⁴, transfer-learning PINNs¹⁷, Butler–Volmer-guided networks¹⁸, temperature-coupled PINNs¹⁹, and membrane-resistance-based hybrid models²⁰ also highlight the rapid growth of physics-guided approaches.

Learning from these various approaches, the present work proposes an LSTM–PINN that uses Arrhenius-based impedance growth behaviour and temperature-dependent degradation kinetics as physical constraints.

Department of Information Science and Engineering, B.M.S. College of Engineering, P.O. Box No.: 1908, Bull Temple Road, Bengaluru, KA 560019, India. email: rashmikh.ise@bmsce.ac.in

This offers improved physical consistency, reduces long-term prediction drift and better generalization across varying cycling conditions.

The recent research in lithium-ion battery is focused more on hybrid architectures that combines the strengths of deep learning with the physical consistency of electrochemical aging laws. As summarized in Table 1, one of the earlier hybrid approaches was the LSTM–PINN framework introduced by Lixin et al.¹⁵, which demonstrated the benefit of integrating temporal sequence modelling with physics-guided regularization. But this method relied on simplified degradation assumptions and showed instability during longer cycles prediction.

Wang et al.¹¹ focused on designing a PINN which extracts statistical descriptors from CC–CV charging profiles thus achieving high accuracy across multiple datasets. But still, as indicated in Table 1, its limitation is to adapt to general cycling conditions because it depends on specific engineered features. Ma et al.⁵ addressed the data scarcity and presented a small-sample PINN formulation. But the lack of monotonicity enforcement allowed occasional non-physical SOH recovery.

More physics focused frameworks have also been explored. Singh et al.¹⁴ embedded a Single-Particle Model (SPM) within a PINN. Chen et al.⁷ and Yang et al.¹⁰ proposed multi-task PINNs that are capable of estimating multiple health indicators. As shown in Table 1, these methods achieved an improved understanding and comprehension of physical laws but require detailed electrochemical parameters. This increases the demand for computation and affects the feasibility of current battery management systems.

Some PINNs have been used for inference-oriented work, such as the electrode-level model by Xiong et al.¹², and the spacecraft-oriented PINN developed by Fu et al.²¹. These examples further outline the trend of using physical laws to constrain the outputs of neural architectures. However, as shown in Table 1, the computational requirements and data availability remain a concern.

These studies reveal two persistent gaps: (1) long-horizon instability in data-driven models that lack strong monotonic physics constraints, and (2) limited practicality of physics-heavy PINN approaches that rely on detailed electrochemical parameterization, complex PDE formulations, or specialized sensing. Motivated by these gaps, the present work adopts a lightweight LSTM–PINN architecture in the setting of offline, laboratory-supervised SOH trajectory forecasting, where SOH values from the initial cycling period are available from capacity-based measurements and are used as historical inputs to predict the subsequent evolution of battery ageing. The Arrhenius- based temperature acceleration and impedance-growth effects have been integrated into the framework, which realistically represents degradation constraints physically such that it is known that better long-horizon predictability, physical feasibility and interpretability of SOH prediction arises in controlled cycling conditions, without the complexity and deployment challenges of parameter-intensive physics model costs. This is unique, in that computationally costly electrochemical solvers are substituted with a lightweight, neural approximation of impedance factors, which modulates Arrhenius degradation law, and eliminates the computational cost of partial differential equations (PDE).

Results

This part outlines the performances of the Proposed Hybrid Enhanced LSTM–PINN model, which confirms its superiority to the traditional CNN and pure LSTM model, which validates the importance of keeping physical parameters consistent during the learning process.

Quantitative performance comparison

The enhanced LSTM–PINN was tested on an unseen test set and compared to the existing literature on SOH estimation models. The findings indicate that physics constraints cause a dramatic improvement in accuracy and stability. Two major benefits of the model are validating over the entire range of the operational battery and improving physical coherence as embodied by the Arrhenius and impedance factors.

The following comparison in Table 2 aims at measuring the robustness and the ability of eneralization of the model over purely data-based techniques and other competing physics-based techniques.

Prior work	Contribution	Limitation	Relevance to this work
Lixin et al. ¹⁵	LSTM–PINN hybrid for SOH and RUL forecasting	Simplified degradation modelling; unstable during long-horizon rollouts	Our model adds Arrhenius + impedance-based monotonic constraints for stable degradation
Wang et al. ¹¹	PINN using CC–CV statistical descriptors across datasets	Protocol-dependent engineered features limit generalization across cycling patterns	We use intrinsic degradation laws independent of cycling protocols
Ma et al. ⁵	Small-sample PINN formulation for SOH estimation	No monotonicity enforcement; occasional non-physical SOH recovery	We enforce strict monotonic degradational behavior for physical consistency
Singh et al. ¹⁴	PINN integrated with Single Particle Model (SPM) for SOC/SOH estimation	Requires detailed electrochemical parameters and PDE computation	Our approach avoids PDE physics and depends only on measurable cycling features
Chen et al. ⁷	Small-sample PINN with physics-guided regularization	Requires partial electrochemical parameterization	Our method uses explicit Arrhenius + impedance laws for clear degradation interpretation
Yang et al. ¹⁰	Multi-task PINN for capacity, resistance, and internal-state estimation	Requires multi-channel sensor data and detailed labels	Our model relies only on standard cycling measurements (SOH, <i>T</i> , impedance)
Xiong et al. ¹²	Electrode-level PINN diagnostics	Relies on morphology-specific or chemistry-specific calibration	Our method remains chemistry-agnostic and morphology-independent
Fu et al. ²¹	PINN for spacecraft battery prognostics	Computationally heavy and unsuitable for embedded BMS	Our framework is lightweight and deployable in real-time BMS

Table 1. Comparison of key prior works on physics-informed battery prognostics.

Model	Dataset	RMSE (%)	MAE (%)	R ²
Proposed LSTM–PINN	NASA battery cycling dataset	1.01	0.62	0.9863
Pure LSTM baseline	NASA battery cycling dataset	4.31	4.11	0.8674
PINN (general, small sample) ⁷	NASA battery cycling dataset	1.85	1.45	0.9650
PINN (membrane resistance) ²⁰	NASA battery cycling dataset	1.58	1.15	0.9710
Hybrid PINN–LSTM framework ¹⁵	NASA battery cycling dataset	2.15	1.70	0.9585
Standard CNN (Data-driven) ¹	NASA battery cycling dataset	2.50	2.05	0.9400

Table 2. Performance comparison between proposed LSTM–PINN model and literature baselines. Significant values are in bold.

Model Variant	Prediction behaviour	Long-horizon stability	Qualitative observation
Full LSTM–PINN (all physics terms)	RMSE ≈ 1.01% (reference baseline); trajectories closely follow ageing trend	Stable across roll-out	Physically coherent and monotonic degradation trajectory
Monotonicity Constraint Removed	Accuracy comparable in short horizon	Less stable; occasional deviations	Local recovery-like bumps and oscillations appear
Impedance-Growth Constraint Reduced	Moderately worse than baseline in ageing regions	Drift increases over longer horizons	Larger deviation in segments with higher impedance evolution
Arrhenius Temperature Term Reduced	Comparable under nominal conditions	Mild degradation at elevated temperatures	Tends to underestimate ageing rate at higher temperature

Table 3. Qualitative sensitivity analysis of physics-informed constraints.

A qualitative sensitivity inspection was carried out to examine the contribution of each physics-informed constraint to the prediction behaviour, as summarised in Table 3. The full LSTM–PINN model (reference baseline; RMSE ≈ 1.01%) produced the most stable and physically consistent SOH trajectories during long-horizon roll-outs. When the monotonicity constraint was removed, the predictions remained generally accurate but exhibited occasional recovery-like artefacts, indicating a loss of physical plausibility in certain regions of the trajectory. Reducing the impedance-growth constraint led to a noticeable increase in long-horizon drift, particularly in ageing segments associated with higher impedance evolution. Weakening the Arrhenius temperature-acceleration term had a milder effect, with performance remaining similar under nominal conditions but tending to underestimate the ageing rate at elevated temperatures. Taken together, these observations indicate that the three physics components contribute in complementary ways - monotonicity improves physical coherence, the impedance term stabilises long-range trajectory behaviour, and the Arrhenius term reinforces temperature-dependent ageing trends. The high fidelity of the hybrid model is evidenced by the correlation between predicted and actual SOH values (Fig. 1), yielding an R^2 coefficient of 0.9863 and confirming the model’s ability to track battery degradation across a wide range of health states.

Analysis of prediction trajectory and stability

The ability of the model to adhere to complex and non-linear change directly reflects on the performance curves. The proposed model prediction curve is in agreement with the actual SOH pattern to a great extent, throughout the battery life. We observe a low Mean Absolute Error Percentage (MAPE = 0.83%). However, the test metrics contained a Maximum absolute error (Max Error) value of 8.38% SOH. So, these extreme cases are infrequent but they bring evidence of the challenge posed by sensor noise or anomalies and essentially inexplicable decay events. However, the challenges notwithstanding, the building of this model ensures high average accuracy.

- 1. *Monotonicity Enforcement:* The Low MAE and High R^2 show that the $\mathcal{L}_{mono}$ (Monotonicity) loss is successful and in most instances, non-physical SOH recovery (i.e. recovery that is anticipated) is prevented (peacefully) when it has not been observed in pure data-driven models.
- 2. *Robustness:* robust regression loss functions (Huber, SmoothL1) are resilient to the outliers that bias the Maximum Error, such that the model is concerned with the overall physical trend.

Analysis of prediction outliers

An analysis of the worst-case prediction samples (Maximum Error ≈ 8.38%) reveals that extreme errors are predominantly concentrated at “capacity regeneration” events. These examples are instances where the state of health (SOH) has sudden upward spikes can be attributed to periods of battery rest.

Failure cause

The model applies a very strict constraint of monotonicity ($\mathcal{L}_{mono}$) in order to maintain physical plausibility as defined in the training configuration. In cases where the ground-truth level is in violation of the physical degradation laws i.e. when the non-physical capacity restoration is reflected in the data, the physics-constrained model will appropriately reject such non-physical signals hence maintaining steady downward trend. Therefore, the quantitative error will have spikey features locally, although the prognostic performance of such a model will be higher than a model that overfits these anomalies mindlessly.

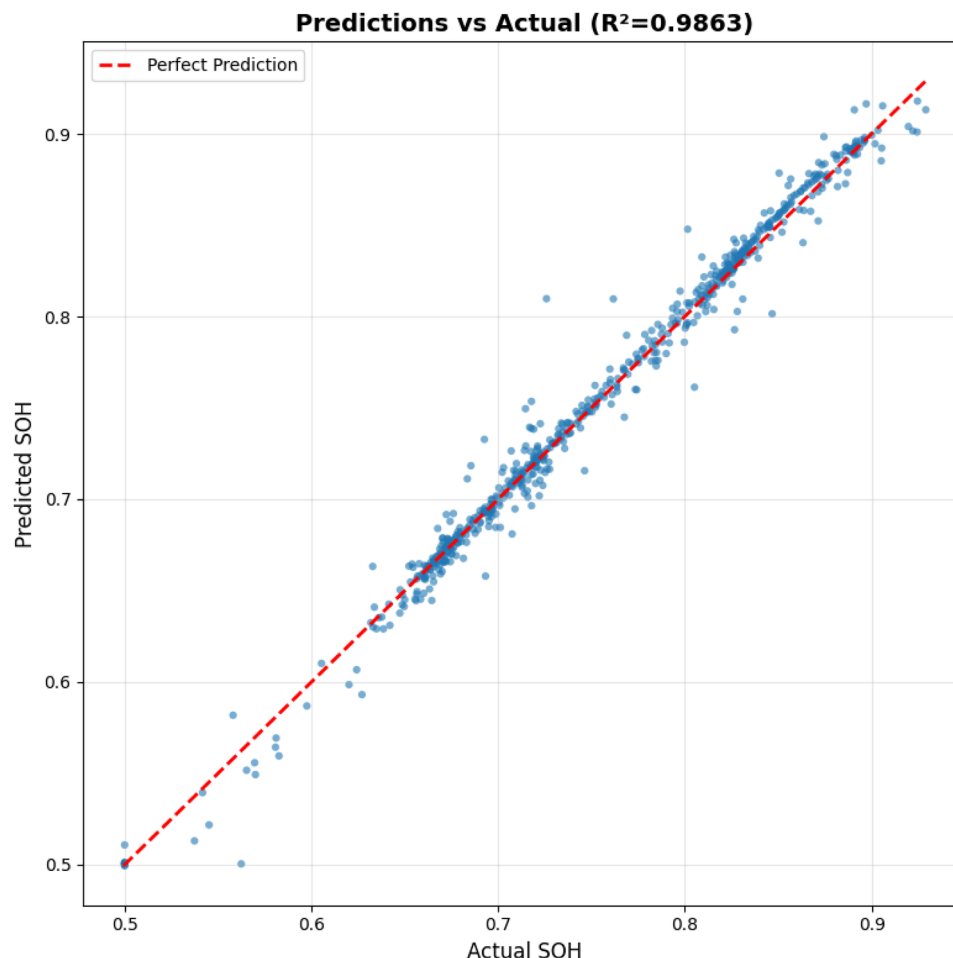


Fig. 1. Distribution of predicted SOH versus actual SOH.

Proposed remedy

While robust loss functions (Huber Loss) partially mitigate the gradient impact of these outliers, future iterations could incorporate a term into the governing equation to explicitly model reversible capacity gain, allowing the physics head to distinguish between sensor noise and genuine relaxation effects.

Discussion

One of the strengths of the LSTM-PINN structure is the capacity to learn electrochemical and physical parameters. The analysis of the learned parameters using the test set are very consistent, according to the summary report:

1. **Activation energy (E_a):** The E_a value of 39.98 ± 0.004 kJ/mol that has been learned is consistent with the processes of degradation chemically, e.g., formation and growth of SEI layers (literature range is usually 20–80 kJ/mol). This brings it to first-hand confirmation that the Arrhenius of the governing equation (2) was effective in bringing the network to generalized, degradation mechanism on the basis of physics.
2. **Model convergence:** All learned physics parameter (α , β and E_a) have a negligible standard deviation (± 0.00) throughout the test set, indicating that the model has reduced to one robust physical law. That is a first-hand testimony to the performance of the Bayesian Optimization technique in the reduction of the physical residual.
3. **Impedance factor impact:** It is the model which has better performance in terms of accuracy than other high- R^2 baselines (see Table 2) due to the non-linear Neural Impedance Factor (F_{imp}). This aspect was acquired, taught by the dedicated DeepHTPM branch (3), which enables the model to dynamically regulate the degradation rate on the basis of real-time external and internal factors, which is where the traditional models, such as the simple PINN tend to fail.

This study presented an enhanced LSTM-PINN hybrid framework for lithium-ion battery State of Health (SOH) estimation, integrating an LSTM-based degradation predictor with physics-informed constraints derived from electrochemical ageing behaviour. The proposed model addresses key limitations of purely data-driven approaches, particularly drift, instability and lack of physical coherence in long-horizon SOH predictions, which

have been reported in prior literature^{1,7,15}. The framework achieved an RMSE of 1.01%, MAE of 0.62% and R^2 of 0.9863 on the NASA Battery Cycling Dataset, and produced smoother and more physically consistent degradation trajectories compared to the baseline data-driven models.

The observed improvement in prediction stability is attributed to the incorporation of Arrhenius-based temperature dependence and the impedance-linked degradation factor in the physics head, together with the monotonicity constraint and the PDE-residual penalty, which together act as physics-guided regularisers. The learned activation-energy term remained within physically reasonable bounds and was consistent with ranges reported in degradation and impedance-growth studies^{9,10}, supporting the physical plausibility of the resulting trajectory behaviour within the evaluated setting.

The evaluation in this work is intentionally conducted under a controlled, well-characterised ageing dataset, enabling a focused examination of the predictive behaviour and stability properties of the proposed framework without confounding variation arising from heterogeneous chemistries or usage regimes. Accordingly, the study is positioned as a methodological investigation into the benefits of physics-guided constraints for SOH trajectory forecasting, rather than as a large-scale benchmarking exercise across multiple datasets.

Statistical analysis of the residual distribution (Fig. 2) reveals a zero-centered error frequency, suggesting that the physics-informed constraints prevent systematic bias even as the battery approaches the end-of-life regime.

Future work will extend the framework toward long-horizon Remaining Useful Life (RUL) prediction with improved robustness under extended roll-out, building on recent developments in RUL-oriented hybrid prognostics^{15,16}. Further directions include the incorporation of uncertainty quantification for reliability assessment in low-SOH regimes, for example via Monte Carlo dropout or Bayesian formulations¹⁵, and the investigation of transfer learning strategies to enable reuse of physics-guided parameters across cells with limited historical data¹⁶. In addition, the framework may be extended toward multi-state co-estimation, including potential joint estimation of SOH with SOC or internal temperature states, as explored in recent electrochemical-informed modelling efforts^{9,10}.

Methods

The proposed method introduces a hybrid physics-driven neural network architecture which is designed to accurately estimate the State of Health (SOH) of lithium-ion batteries. Data-driven approaches like LSTMs effectively captures time dependencies¹ but outside the training distribution, they often lack generalization. To address this, our scheme integrates the physics of battery degradation into the learning process. This approach combines empirical time-series data with thermodynamic and electrochemical laws. This is similar to the frameworks explored in² and¹⁶. The model is trained to learn degradation patterns from historical data but it also ensures that these predictions matches the known physical equations. This prevents the output from violating the actual physical behavior of the battery⁶.

The approach is that the system models battery degradation as a first-order differential equation which solves the rate of health decay as a function of internal electrochemical conditions. The interactive effects of ambient temperature and impedance evolution is aggregated by this differential equation, which determines the effective degradation rate. The governing parameters captures the acceleration of aging because of the elevated temperatures¹⁹ and the growth of internal resistance^{12,20}. These represent the primary degradation mechanisms in lithium-ion batteries.

Governing degradation model

Battery ageing is modelled as a first-order degradation process in which the State of Health (SOH) decreases monotonically with cycling and operating stress. Rather than treating ageing as a purely empirical curve-fit, the degradation trajectory is represented through a physics-consistent parametric formulation that captures slow early ageing followed by accelerated decay at later stages, consistent with physics-informed battery ageing models^{13,16}.

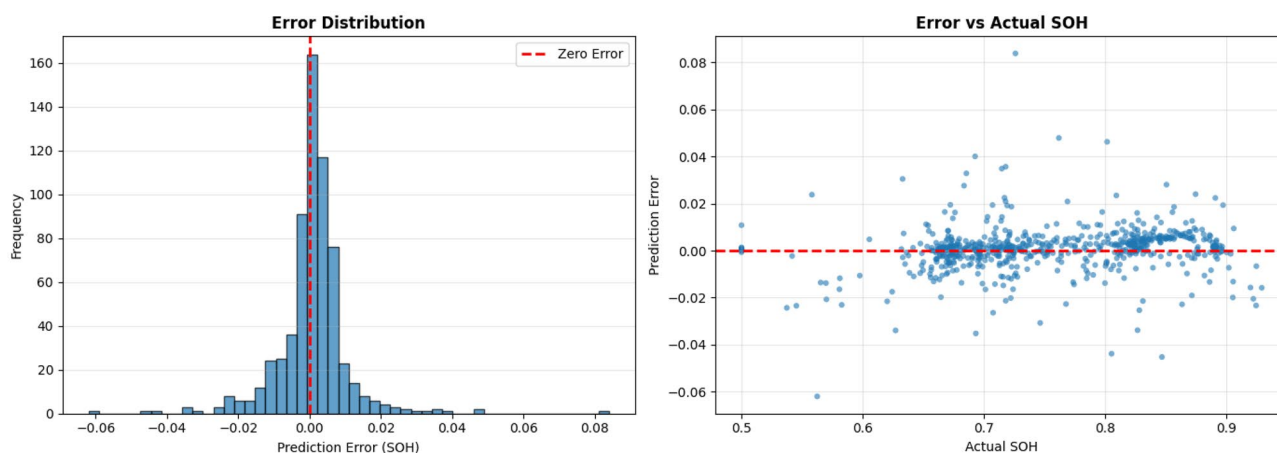


Fig. 2. Analysis of predicted error versus actual error.

The degradation dynamics are modelled as

$$\frac{d SOH}{dt}_{\text{physics}} = -\alpha_{\text{eff}}^{\beta} (1 - \alpha_{\text{eff}} t)^{\beta-1}, \quad (1)$$

where

- SOH is the State of Health of the battery;
- t is the normalised cycle index;
- α_{eff} is the effective degradation-rate coefficient, which determines the overall rate of capacity loss per cycle; and
- β is the ageing-shape parameter, which controls the curvature of the degradation trajectory ($\beta < 1$ gradual early-life ageing, $\beta \approx 1$ near-linear decay, $\beta > 1$ accelerated late-life ageing).

This formulation represents a monotonic and progressively accelerating ageing behaviour that is consistent with observed lithium-ion battery degradation.

The effective degradation coefficient α_{eff} is not treated as a fixed constant; instead, it accounts for thermally accelerated ageing and impedance-driven degradation, and is defined as

$$\alpha_{\text{eff}} = \alpha_{\text{base}} \exp\left(\frac{E_a}{R} \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T}\right)\right) \cdot F_{\text{imp}}(\Delta R_e, \Delta R_{ct}), \quad (2)$$

where

- α_{base} is the nominal reference degradation-scaling coefficient;
- E_a is the activation energy associated with temperature-accelerated ageing;
- R is the universal gas constant;
- T is the operating temperature and T_{ref} is the reference temperature;
- $F_{\text{imp}}(\Delta R_e, \Delta R_{ct})$ is the impedance growth factor capturing the effect of electrolyte resistance change ΔR_e and charge-transfer resistance change ΔR_{ct} on ageing.

The Arrhenius term models thermally induced acceleration of degradation¹⁹, while the impedance factor $F_{\text{imp}}(\Delta R_e, \Delta R_{ct})$ represents degradation arising from resistance growth mechanisms that are widely reported as dominant contributors to lithium-ion battery ageing^{12,20}.

Rather than prescribing a fixed analytical impedance law, $F_{\text{imp}}(\cdot)$ is approximated using a compact neural sub-network, enabling the model to represent nonlinear electrochemical interactions while remaining constrained by the governing physical structure. The parameter α_{base} remains fixed during training and serves as a physical reference scale, ensuring that temperature effects are interpreted relative to a meaningful baseline rather than being absorbed into optimisation-driven curve fitting.

The parameters α_{eff} and β therefore play complementary but physically distinct roles in the degradation model. The coefficient α_{eff} governs the *overall rate* at which SOH decreases under a given operating condition, and increases in response to thermally and impedance-induced stress. Conversely, the parameter β controls the time-dependent evolution of battery aging, dictating whether degradation progresses gradually or transitions into a significantly accelerated late-life regime. Taken together, the effective parameter α_{eff} establishes the characteristic timescale of capacity fade, while β defines the aging pathway. This enables the model to describe both the monotonic decay of performance and the experimentally observed acceleration in degradation during later cycles.

Hybrid neural architecture

The framework proposed follows a hybrid schematic design (two-branch) where ageing tendencies of the temporal nature are derived based on the operational history and the ensuing SOH trajectory is dictated by the degradation law expressed in (1).

The former branch includes a Long Short-Term Memory (LSTM) encoder that takes past SOH sequences that are coincident with the operating variable in question. The LSTM compresses a latent state that captures long-term time dynamics and ageing processes thus providing a small embedding of the observed degradation dynamics^{1,2,16}.

The second branch is a physics-informed head inspired by hybrid PINN formulations^{8,15}. This head maps the latent representation together with the current temperature, electrolyte resistance, and charge-transfer resistance - to the parameters governing the degradation model, namely β , E_a , and the impedance contribution to α_{eff} .

The temporal encoder is a two-layer LSTM (hidden size = 64, dropout = 0.1), whose final hidden state forms the latent trajectory representation. The data-driven SOH head is a two-layer MLP (64→64→1) with ReLU and Dropout(0.2) followed by a Sigmoid output constraint, while the physics head is a three-layer Tanh MLP (128 hidden units) that outputs α_{base} , β , and E_a , each constrained to conservative literature-consistent ranges.

The integrated architecture of the hybrid model is illustrated in Fig. 3, detailing the concurrent processing of SOH history via LSTM layers and the derivation of physical parameters- α , β , and E_a -through the DeepHTPM branch.

In this formulation, the latent time-series features condition the *physical parameters* rather than directly outputting SOH values. The predicted parameters are further used to solve the differential equation in (1), hence producing a simultaneously data-adapted and physically consistent degradation trajectory. This combination

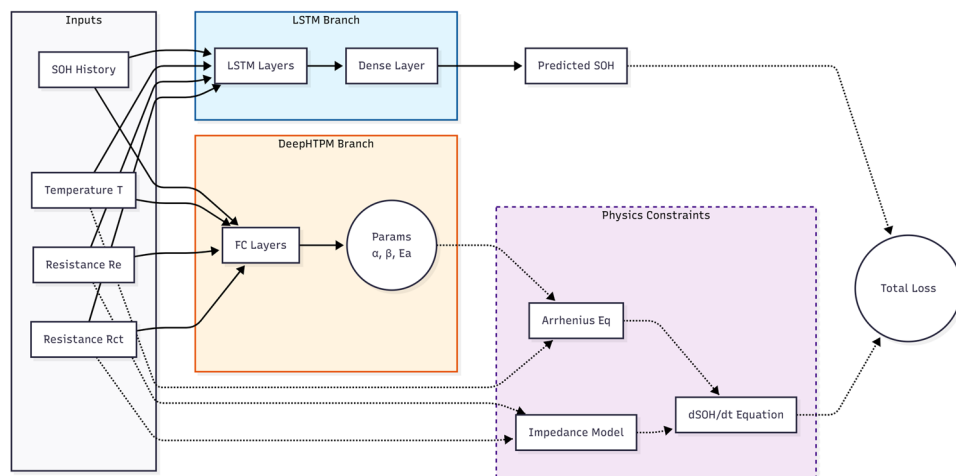


Fig. 3. Proposed hybrid LSTM-PINN architecture.

also allows the temporal variability to be learned by the LSTM and rules out non-physical events such as state-of-health healing or physically implausible decay in the end of the cycle.

Physics-consistent compound loss function

The model training is presented as a multi-objective optimization problem that allows balancing the accuracy of the reconstruction and empirical strain on the compliance of the law of degradation. The total loss is expressed as

$$L_{total} = \lambda_{pred} L_{pred} + \lambda_{pde} L_{pde} + \lambda_{mono} L_{mono} \quad (3)$$

The prediction loss L_{pred} enforces agreement between predicted and measured SOH values and is implemented using the Huber loss to improve robustness under noise and measurement artefacts^{5,7}.

The physics-informed term L_{pde} penalises deviations between the data-derived temporal gradient and the physics-derived gradient from (1). The residual is computed at collocation points as

$$f(t; \theta) = \frac{\Delta SOH}{\Delta t} - \frac{dSOH}{dt}_{physics} \quad (4)$$

and evaluated using a Smooth L1 penalty to stabilise the magnitude of residual gradients during optimisation^{10,22}. This term constrains the network to remain consistent with the physically derived degradation dynamics, rather than relying solely on data-driven fitting.

Finally, the monotonicity constraint L_{mono} penalises instances in which SOH increases between successive cycles, suppressing non-physical recovery behaviour and ensuring strictly degrading dynamics across time⁹.

Together, these loss components encourage the model to produce SOH trajectories that match empirical observations while preserving thermodynamic and electrochemical plausibility as encoded in the governing degradation model.

Network optimisation, hyperparameter search, and training configuration

The optimisation problem is formulated as finding the parameter set θ^* that minimises the compound loss function:

$$\theta^* = \arg \min_{\theta} L_{total}(\theta) \quad (5)$$

where the gradients are computed jointly from the prediction error and the residual of the underlying physical degradation law²². The Adam optimiser is employed due to its adaptive learning-rate behaviour and implicit momentum characteristics, which enable stable and efficient convergence in a high-dimensional parameter space. This is particularly important for the hybrid learning setting, where the model simultaneously fits empirical SOH trajectories while satisfying physics-derived constraints.

To further improve convergence behaviour and parameter sensitivity, Bayesian optimisation¹⁶ is used to automatically tune key hyperparameters. The search space includes the learning rate, the weighting coefficients for the physics-consistency and monotonicity terms (λ_{pde} , λ_{mono}), the number of hidden units in the network, and the batch size. The Tree-structured Parzen Estimator (TPE) algorithm is adopted to explore the hyperparameter space efficiently, prioritising high-promise regions and yielding substantially improved sampling efficiency compared to grid or random search. The optimisation objective is strictly defined as the minimisation of the validation loss L_{total} across a bounded set of trials, enabling the selection of the configuration that generalises best to unseen battery instances.

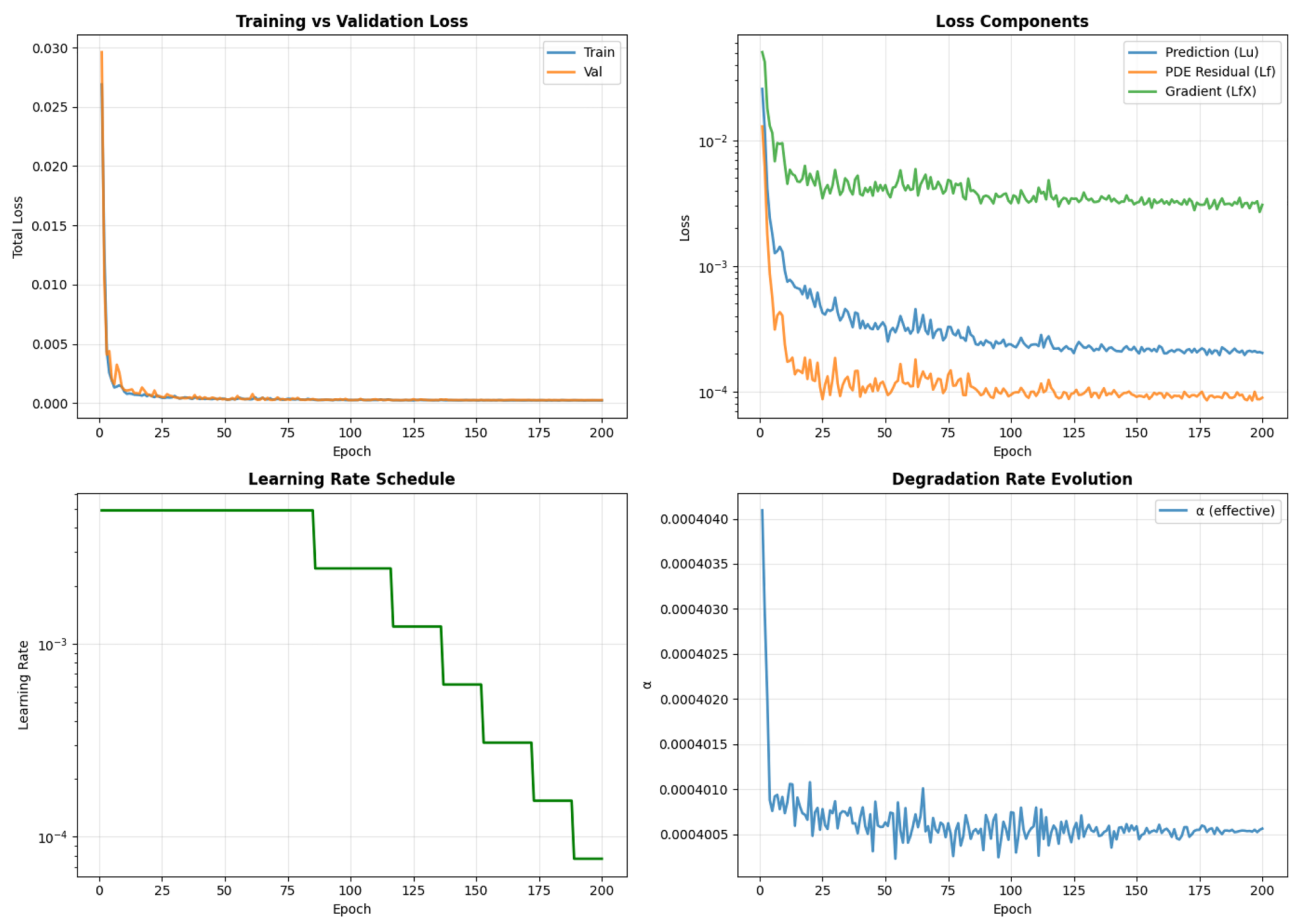


Fig. 4. Training dynamics and Parameter evolution.

Training is conducted over a fixed number of epochs, during which the model simultaneously optimises two complementary objectives: (1) correspondence with empirical SOH measurements and (2) consistency with the physics-surface runaway constraint applied at collocation points²³. To prevent overfitting and ensure generalisation, checkpointing is employed to retain the parameter set that minimises the validation loss. The active learning-rate schedule follows the performance of a validation range and cuts down on the step size when the measure has leveled off, which allows ultrafine-scale adjustments of weights in later convergence cycles.

Model convergence and the optimization of individual loss components are shown in Fig. 4; notably, the simultaneous decline of the prediction loss (L_u) and the physics-based residual (L_f) demonstrates that the network effectively learned the data distribution while adhering to the governing degradation laws.

The last model was trained using 200 epochs and the hyperparameters were determined using Bayesian search on a 20% validation split. The resultant structure was tested on new operating regimes unattainable before to determine the ability of inter-battery generalisation. Combining empirical and physics optimisation, the integrated empirical and physics optimisation guarantees that the inferred degradation models have statistical fidelity to the measured SOH trends as well as a physically feasible behaviour in a wide range of temperature and history of impedance histories.

The model was implemented using the PyTorch deep learning framework.

- **Hardware:** A significant amount of GPU resources were used, i.e. the `pin_memory` option was used in order to be able to load various data well.
- **Initial configuration:** The default training configuration parameters were set as Rated Capacity 2.0 Ah, Epochs 200 and initial Batch Size 32 prior to the optimization.
- **Losses:** The initial coefficients of weights for our loss function were set as $\lambda_{\text{pred}} = 1.0$, $\lambda_{\text{pde}} = 0.1$, and $\lambda_{\text{mono}} = 0.01$. These weights were optimized using Bayesian Optimization using the OPTUNA library.

Dataset description and splitting

The model was evaluated using the NASA Li-ion Battery Aging Dataset. This dataset provides a metadata, including State of Health (SOH) capacity measurements, ambient temperature and derived impedance parameters Electrolyte Resistance (R_e) and Charge Transfer Resistance (R_{ct}) for 34 battery samples.

An inter-battery splitting strategy was employed to validate the model's generalization capabilities across different battery variants, initial states and varying conditions. The total set of 34 batteries was randomly shuffled (using a fixed random seed of 42) and partitioned by Battery ID into three mutually exclusive sets:

- *Training set*: 70% of the batteries (These sets are used for fitting scalars and model training).
- *Validation set*: 15% of the batteries (These sets are used for monitoring loss during training and selecting optimal hyperparameters).
- *Test set*: 15% of the batteries (These sets are held out completely for final performance evaluation).

Only batteries possessing a minimum of 50 cycles were included in the final dataset for sequence creation.

Data preprocessing

The preprocessing pipeline operates on discharge-cycle records from the NASA Li-ion Battery Aging Dataset. The State of Health (SOH) is computed from discharge capacity relative to the rated capacity and subsequently smoothed using a Savitzky–Golay filter (window length = 15, polynomial order = 2), which suppresses high-frequency measurement noise while preserving the underlying degradation trend and the derivative shape required for stable computation of the physics-informed loss.

The electrolyte resistance (R_e) and charge-transfer resistance (R_{ct}) used in this work are taken directly from the impedance-derived diagnostic parameters supplied by the dataset, rather than computed or refitted within our pipeline. Missing impedance samples are handled using forward- and backward-filling to maintain temporal continuity, and remaining invalid values are replaced with small nominal defaults to prevent numerical singularities in the impedance-growth term. As these quantities originate from the dataset's impedance-processing pipeline, any associated fitting uncertainty is inherent to the dataset rather than introduced by this study; within our framework they are treated as externally supplied diagnostic indicators of impedance evolution.

Feature normalisation for temperature, R_e , and R_{ct} is performed using standard-score scaling, with scalars fitted exclusively on the training set to avoid information leakage into validation and test partitions. Model input sequences are then constructed using a sliding-window procedure, where a length- L historical window forms the input and the subsequent SOH value constitutes the prediction target.

Sequence creation and input features

The model's time dependent values are captured using a sliding window approach. This defines the input historical length ($\mathcal{L}$).

- *Sequence length*: 20 cycles input window size ($\mathcal{L}$) was used to build the input sequences (SOH_{history}). An empirical sensitivity analysis that was studied in various candidate window lengths (5, 10, 20, 30, and 40 cycles) was used to select the window length. This assessment was informed by parameters, including the prediction accuracy, stability at long horizons, as well as, noise robustness. Very short windows (e.g. 5–10 cycles) did not give enough time horizon context to age, because the dynamics of state-of-health (SOH) degradation are slow and smooth, then unstable roll-out behavior occurs in longer horizon predictions. On the other hand, larger window size (≥ 30 cycles) added redundancy to the input, model variance, and reduced generalization among the shorter available histories in the battery. The 20 cycle window was a good tradeoff between the ability to capture temporal dependencies and the ability to keep the model robust enough, providing sufficient information about degradation evolution to identify initial monotonic trends, but not being over-trained in longer sequences. Such a structure also augers well with such practical aspects of the studies like availability of data sets and feasible observation horizons in the case of laboratory cycling studies.
- *Feature Inputs*: The model accepts four simultaneous inputs which include SOH, Temperature, Electrolyte Resistance (R_e), and Charge Transfer Resistance (R_{ct}).

Data normalization and smoothing

1. *Target Smoothing*: The SOH target values and input sequences were smoothed using a Savitzky–Golay filter (window length = 15, polynomial order = 2) to suppress high-frequency measurement noise while preserving the underlying degradation trend and derivative shape. This preprocessing is applied in an *offline laboratory setting* and is used to obtain physically coherent SOH trajectories for training. The smoothing improves the stability of the monotonic degradation behaviour and enables robust computation of the physics-informed loss term (L_{pde}), which relies on gradient consistency rather than pointwise interpolation¹³. This smoothing is not intended to represent an online filtering operation.
2. *Feature Normalization*: All additional physics features (T , R_e , R_{ct}) were scaled using the StandardScaler (Z-score normalization). These scalars were fitted exclusively on the training set to prevent data leakage into the validation and test sets.
3. *Data Handling*: Explicit protection was provided in the preprocessing to impute the missing value of impedance by forward-filling and backwards-filling techniques and to give a non-zero initialisation of the R_e and R_{ct} to preclude the mathematical singularities in the calculation of the impedance growth factor (Equation 2).

Data availability

The dataset analysed during the current study is available on [nasa.gov](https://phm-datasets.s3.amazonaws.com/NASA/5.+Battery+Data+Set.zip). **Data Set Citation** : B. Saha and K. Goebel (2007). "Battery Data Set", NASA Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA.

Code availability

Code supporting this study is available on GitHub: [GitHub Repository](#).

Received: 3 December 2025; Accepted: 27 January 2026

Published online: 06 February 2026

References

1. Rincón-Maya, C., Guevara-Carazas, F., Hernández-Barajas, F., Patino-Rodriguez, C. & Usuga-Manco, O. Remaining useful life prediction of lithium-ion battery using ICC–CNN–LSTM methodology. *Energies* **16**, 7081. <https://doi.org/10.3390/en16207081> (2023).
2. Ding, Y., Zhu, J., Liu, Y., Ning, D. & Qin, M. Integrated framework of LSTM and physical-informed neural network for lithium-ion battery degradation modeling and prediction. *AI* **6**, 149. <https://doi.org/10.3390/ai6070149> (2025).
3. Xue, C., Jiang, B., Zhu, J., Wei, X. & Dai, H. An enhanced single-particle model using a physics-informed neural network considering electrolyte dynamics for lithium-ion batteries. *Batteries* **9**, 511. <https://doi.org/10.3390/batteries9100511> (2023).
4. Wang, Y. et al. Physics-informed recurrent neural networks with fractional-order constraints for the state estimation of Lithium-Ion batteries. *Batteries* **8**, 148. <https://doi.org/10.3390/batteries8100148> (2022).
5. Ma, J. et al. Physics-informed neural soh estimation method for lithium-ion battery under partial observability and sparse sensor data. *Research Square*. <https://doi.org/10.21203/rs.3.rs-6806872/v1> (2025). Preprint.
6. Deng, Y., Du, C. & Ren, Z. A novel method for estimating the state of health of lithium-ion batteries based on physics-informed neural network. *Batteries* **11**, 49. <https://doi.org/10.3390/batteries11020049> (2025).
7. Chen, L. et al. Physics-informed neural networks for small sample state of health estimation of lithium-ion batteries. *J. Energy Storage* **122**, 116559. <https://doi.org/10.1016/j.est.2025.116559> (2025).
8. Sun, X., Wei, Z., Li, Y., Lu, H. & Liu, C. New insights into estimating lithium-ion battery cell health leveraging physics-informed machine learning. *IEEE Trans. Instrum. Meas.* **74**, 1–15. <https://doi.org/10.1109/TIM.2025.3575975> (2025).
9. Deng, W. et al. Generic physics-informed machine learning framework for battery remaining useful life prediction using small early-stage lifecycle data. *SSRN Electron. J.* <https://doi.org/10.2139/ssrn.4770354> (2024).
10. Yang, L., He, M., Ren, Y., Gao, B. & Qi, H. Physics-informed neural network for co-estimation of state of health, remaining useful life, and short-term degradation path in Lithium-Ion batteries. *Appl. Energy* **398**, 126427. <https://doi.org/10.1016/j.apenergy.2025.126427> (2025).
11. Wang, Q., Ye, M., Li, B., Lian, G. & Li, Y. Co-estimation of state of charge and capacity for battery packs in real electric vehicles with few representative cells and physics-informed machine learning. *Energy* **306**, 132520. <https://doi.org/10.1016/j.energy.2024.132520> (2024).
12. Xiong, R., He, Y., Sun, Y., Jia, Y. & Shen, W. Enhanced electrode-level diagnostics for lithium-ion battery degradation using physics-informed neural networks. *J. Energy Chem.* <https://doi.org/10.1016/j.jechem.2025.01.019> (2025).
13. Wang, F., Zhai, Z., Zhao, Z., Di, Y. & Chen, X. Physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis. *Nat. Commun.* **15**, 4332. <https://doi.org/10.1038/s41467-024-48779-z> (2024).
14. Singh, S., Ebongue, Y. E., Rezaei, S. & Birke, K. P. Hybrid modeling of lithium-ion battery: Physics-informed neural network for battery state estimation. *Batteries* **9**, 301. <https://doi.org/10.3390/batteries9060301> (2023).
15. Lixin, E. et al. A physics-informed neural network-based method for predicting degradation trajectories and remaining useful life of supercapacitors. *Green Energy Intell. Transp.* **4**, 100291. <https://doi.org/10.1016/j.geits.2025.100291> (2025).
16. Nascimento, R. G., Viana, F. A. C., Corbetta, M. & Kulkarni, C. S. A framework for Li-Ion battery prognosis based on hybrid Bayesian physics-informed neural networks. *Sci. Rep.* **13**, 13856. <https://doi.org/10.1038/s41598-023-33018-0> (2023).
17. Hofmann, T. et al. Physics-informed neural networks for state of health estimation in lithium-ion batteries. *J. Electrochem. Soc.* **170**, 090524. <https://doi.org/10.1149/1945-7111/acf0ef> (2023).
18. Seo, Y., Kim, T. & Barde, S. Enhancing battery SOH prediction with Butler–Volmer informed neural networks in data-scarce environments. *Energy* **335**, 138316. <https://doi.org/10.1016/j.energy.2025.138316> (2025).
19. Shen, K. et al. Physics-informed machine learning estimation of the temperature of large-format lithium-ion batteries under various operating conditions. *Appl. Therm. Eng.* **269**, 126200. <https://doi.org/10.1016/j.applthermaleng.2025.126200> (2025).
20. Luo, Y., Ju, S., Li, P. & Zhang, H. A method for estimating lithium-ion battery state of health based on physics-informed hybrid neural network. *Electrochim. Acta* **525**, 146110. <https://doi.org/10.1016/j.electacta.2025.146110> (2025).
21. Fu, H. et al. Physics-informed neural network for spacecraft lithium-ion battery modeling and health diagnosis. *IEEE/ASME Trans. Mechatron.* **29**, 3546–3555. <https://doi.org/10.1109/TMECH.2023.3348519> (2024).
22. Tian, A. et al. A generic physics-informed neural network framework for lithium-ion batteries state of health estimation. *Energy* **332**, 137215. <https://doi.org/10.1016/j.energy.2025.137215> (2025).
23. Dang, L., Yang, J., Liu, M. & Chen, B. D. Differential equation-informed neural networks for state of charge estimation. *IEEE Trans. Instrum. Meas.* **73**, 1–12. <https://doi.org/10.1109/TIM.2023.3334377> (2024).

Author contributions

P.N.K. developed the methodology and implemented the code. R.K.B served as the corresponding author. P.N.K., P.R.U., and N.S. wrote the original manuscript draft. P.R.U. and P.K.G. collected and validated the data. R.K.B. and S.T.S. provided supervision and technical guidance throughout the work.

Funding

Open access funding provided by B.M.S. College of Engineering.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to R.K.B.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026